

XUESONG BAI

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EDUCATION

University of California, Irvine Ph.D. candidate in Computer Engineering Advisor: Prof. Zhou Li	09/21 - 06/27 (expected)
Chongqing University & University of Cincinnati B.S. in Electrical Engineering	09/15 - 04/20

RESEARCH INTERESTS

My research predominantly focuses on improving the robustness and security of modern network / system infrastructures. Currently, I'm focusing on fuzzing techniques to test network protocol implementations. I have achieved:

- Around 20 CVEs related to a variety of significant DNS vulnerabilities in popular DNS software like ISC BIND 9, and public DNS service providers like Google DNS.
- Papers published in Top-tier conferences like *USENIX SECURITY* and *NDSS*.
- Bug bounty rewards from Google.

SKILLS

Programming Language	C/C++, Python, Golang, Shell Script
Software & Tools	Linux, Git, Docker, Kubernetes, LLVM, GDB, AWS Services, Cloudflare

PROFESSIONAL EXPERIENCES

Software Engineer Intern <i>Meta Platforms, Inc.</i> - Mentor: Yingjie Gu	06/25 - 09/25 Menlo Park, CA
Graduate Student Researcher <i>UC Irvine</i> - Advisor: Professor Zhou Li	06/22 - Present Irvine, CA

SELECTED PROJECTS

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| Configuration-guided Network Service Fuzzing | 2024 - 2025 |
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- Designed *NCFuzz*, a novel fuzzer for discovering bugs hidden behind non-default configurations. The system leverages LLMs to extract configuration semantics from software documentation and applies data-flow analysis to identify configuration-dependent memory locations, enabling guided fuzzing feedback and exploration of code paths reachable only under certain configuration constraints.
 - **Outcome:** Paper currently *in submission*.
 - **Skills:** Fuzzing, Network Configuration, Data Flow Analysis, LLM (Large Language Model).
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| Automatic Cyber-Attack Framework with LLM | 2023 - 2024 |
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- Designed a framework to automate "hands-on-keyboard" attacks on a given network environment, jail-broke and leveraged LLM to analyze the situation and make decisions on the next move. The framework was tested in a simulated organizational network with varied attack tasks, endpoint configurations (Windows and Linux systems) with high success rate.

- **Outcome:** Paper currently *in submission*.
- **Skills:** Network Security, CTF (Capture the Flag), LLM.

Fuzzing on DNS Resolution Implementations

2022 - 2023

- Analyzed 200+ CVEs on DNS software, designed a stateful fuzzing framework with Docker, applied pair-wise network seed mutation, customized scheduler and oracles to find **logic flaws** in the software, used **differential testing** across popular DNS implementations to test resolution consistency.
- 12 types of vulnerabilities, 23 bugs detected, 15 CVEs assigned among 6 DNS implementations.
- **Outcome:** Paper Accepted in Security'24, presented in DNS-OARC'42, NDSS'24 poster session, rewarded by Google Bug Hunters Program.
- **Skills:** Fuzzing, Docker Network Configuration, Software Analysis, Cloudflare API, Python.

DNS Cache Poisoning Attack Analysis & Defense

2021 - 2022

- Investigated and measured the caching mechanism in popular DNS software and public DNS service providers in the global network, found a vulnerability where a revoked domain could still be used for malicious activity for a long time in the current DNS design.
- **Outcome:** 7 DNS software, 15 Public DNS affected; 9 CVEs assigned; Paper accepted in NDSS'23.
- **Skills:** Large-scale Network Measurement, DNS Traffic Analysis, Software Analysis, Python.

PUBLICATIONS

Conference

- Qifan Zhang, **Xuesong Bai**, Xiang Li, Haixin Duan, Qi Li, and Zhou Li. *RESOLVERFUZZ: Automated Discovery of DNS Resolver Vulnerabilities with Query-Response Fuzzing*. In Proceedings of the USENIX Security Symposium (Security), August, 2024. Poster accepted in NDSS'24.
- Xiang Li, Baojun Liu, **Xuesong Bai**, Mingming Zhang, Qifan Zhang, Zhou Li, Haixin Duan, and Qi Li. *Ghost Domain Reloaded: Vulnerable Links in Domain Name Delegation and Revocation*. In Proceedings of the Network and Distributed System Security Symposium (NDSS), February, 2023.

Preprint

- Jiachen Xu, Jack W Stokes, Geoff McDonald, **Xuesong Bai**, David Marshall, Siyue Wang, Adith Swaminathan, and Zhou Li. *Autoattacker: A large language model guided system to implement automatic cyber-attacks*. In submission, available on Arxiv, 2024.

HONORS & AWARDS

- More than 20 CVEs assigned.
- Distinguished Artifact Reviewer Awards, ACM CCS 2024.
- Henry Samueli Endowed Fellowship, 2024.
- Rewards from Google Bug Hunters program, 2022.

ACADEMIC SERVICES

TPC: ICNC (26')

External Reviewer: IEEE S&P (25'), Security ('26), NDSS (24', 25', 26'), ACM CCS (25'), ICNC (25')

Journal Reviewer: Computer Networks, Peer-to-Peer Networking and Applications, IEEE Transactions on Information Forensics & Security

Artifact Evaluation Committee (AEC): NDSS (24', 25', 26'), USENIX Security (24', 25'), ACM CCS (24'), EuroSys (23' Fall).

TEACHING EXPERIENCE

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| Teaching Assistant, EECS 40: OBJ ORIENT SYS & PROG | University of California, Irvine |
| • <i>Instructor: Prof. Zhou Li, Fall 2025 (#students: 126)</i> | |
| Teaching Assistant, EECS 114: ENGR DATA STR & ALG | University of California, Irvine |
| • <i>Instructor: Prof. Yasser Shoukry, Winter 2025 (#students: 98)</i> | |
| Reader, EECS 148 / CompSci 132: Computer Networks | University of California, Irvine |
| • <i>Instructor: Prof. Zhou Li, Spring 2022 (#students: 233)</i> | |
| Teaching Assistant, ENED 3061: PROB STAT I | University of Cincinnati |
| • <i>Instructor: Prof. TJ Murphy, Spring 2020 (#students: 91)</i> | |

Last Updated: May 11, 2026